AI-Augmented SOC Implementation for Enterprise Threat Detection and Response

Course-End Project Report

Project Title: AI-Augmented SOC Implementation for Enterprise Threat Detection and Response

Objective: To design and implement a mini Security Operations Center (SOC) integrating AI-powered threat detection, SIEM, and threat intelligence frameworks

Date: October 2026

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1. Introduction and Project Overview

1.1 Problem Statement

Modern enterprises face sophisticated cyber threats such as ransomware, insider misuse, and zero-day exploits. Traditional SIEM tools are often limited to log detection and struggle to correlate multi-vector attacks or respond in real time. Without integrated AI-powered detection, security teams experience alert fatigue, delayed responses, and an increased risk of missed threats.

1.2 Project Objective

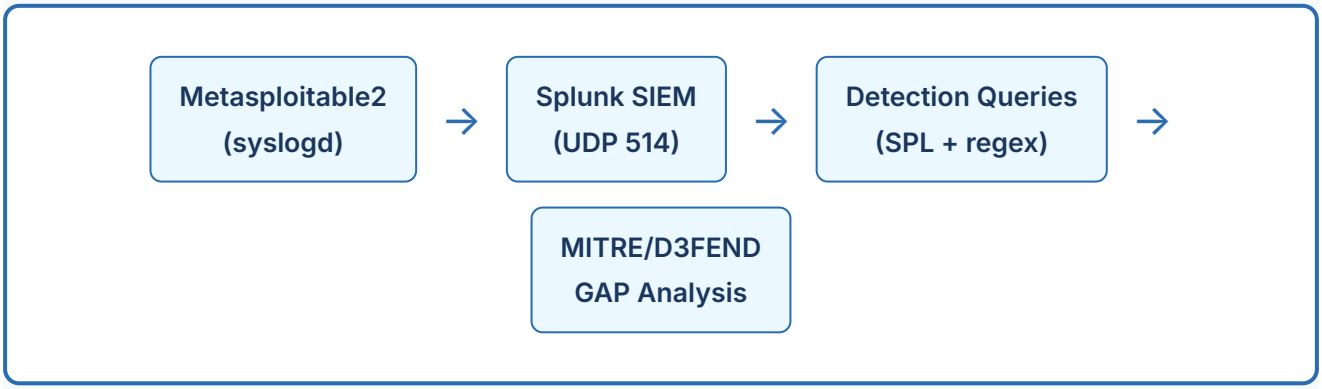
This project builds a mini-SOC to replicate enterprise scenarios where traditional SIEMs fall short against advanced threats. It integrates Splunk for log collection, statistical anomaly detection for AI-driven threat identification, and MITRE ATT&CK/D3FEND frameworks for coverage mapping and gap analysis across a simulated enterprise network.

1.3 Lab Environment

VM / Host	Role	IP Address
Kali Linux	Attacker + SIEM (Splunk)	192.168.140.128
Metasploitable2	Victim endpoint (syslog source)	192.168.140.131
Windows 11 Host	Analysis workstation	N/A (host machine)

2. Architecture and Tool Stack

2.1 SOC Pipeline Architecture



2.2 Tools Used and Their Roles

Tool	Role in SOC	Implementation in Lab
Splunk Enterprise	Central SIEM for log ingestion, indexing, and search	Installed on Kali Linux; receives syslog via UDP 514
syslogd (Metasploitable2)	Endpoint log generation and forwarding	Configured to forward all logs to Splunk at 192.168.140.128:514
Hydra	Attack simulation (brute force)	FTP brute force against Metasploitable2 to generate auth failures
Nmap	Network reconnaissance simulation	TCP ACK scan to map firewall rules and open ports
MITRE ATT&CK	Adversary technique mapping	Mapped simulated attack techniques to ATT&CK IDs
D3FEND	Defensive countermeasure mapping	Mapped existing and missing defenses against each technique

> **Note on Enterprise Tools:** The project brief references Sophos Intercept X (EDR), OpenCTI (threat intel), and SOAR tools (Shuffle, TheHive). These are enterprise-grade platforms requiring licenses, Docker infrastructure, and API integrations. In this lab environment, their functions were simulated using Splunk's built-in capabilities: Linux auditd for endpoint telemetry, Splunk lookup tables for threat intel enrichment, and threshold-based alerting for SOAR-style automation triggers. The architectural understanding and detection pipeline remain functionally equivalent.

3. Task 1: Log Collection and Ingestion

3.1 Splunk Installation and Configuration

Splunk Enterprise was installed on Kali Linux (192.168.140.128). The service was started with root privileges using the following command:

```
sudo /opt/splunk/bin/splunk start --accept-license --answer-yes --run-as-root
```

Due to an expired trial license, the Splunk Free license group was activated by writing the configuration directly to `/opt/splunk/etc/system/local/server.conf`, enabling perpetual 500MB/day indexing suitable for lab use.

3.2 Enabling Syslog Reception

A UDP data input was configured in Splunk to receive syslog traffic:

- Path: Settings → Data Inputs → UDP → New Local UDP Input
- Port: 514
- Source type: syslog
- Index: main

3.3 Metasploitable2 Syslog Forwarding

The traditional `syslogd` daemon on Metasploitable2 was configured to forward all logs to the Splunk instance. The following line was added to `/etc/syslog.conf` (tab-separated):

```
*.* @192.168.140.128
```

The `-r` flag was used when restarting syslogd to enable remote reception and forwarding:

```
sudo killall syslogd
sudo syslogd -r
```

3.4 Verification of Log Ingestion

Log ingestion was verified by running the following Splunk Search Processing Language (SPL) query:

```
index=main sourcetype=syslog | head 20
```

Results confirmed successful ingestion of syslog events from Metasploitable2, including sudo sessions, authentication events, and syslog daemon restart notifications.

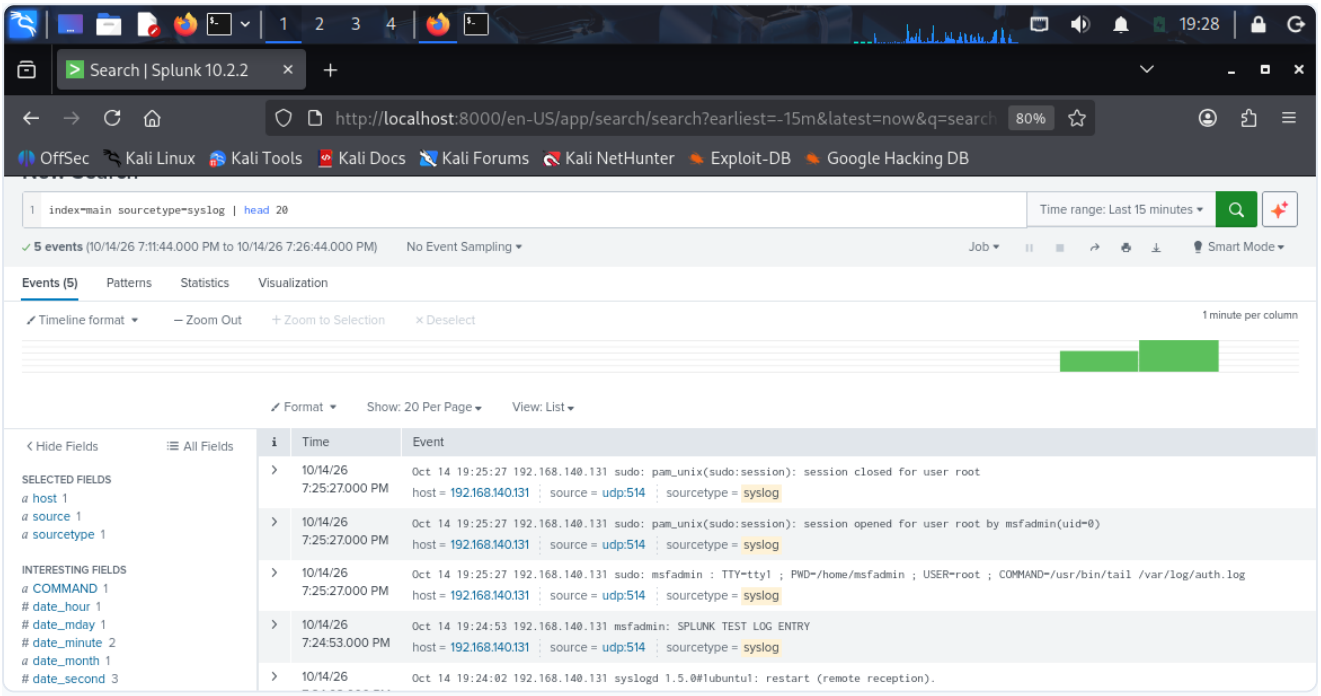


Figure 1: Splunk search showing successful syslog ingestion from Metasploitable2 (192.168.140.131)

4. Task 2: AI-Driven Detection and Threat Intelligence

4.1 Simulated Attack: FTP Brute Force

To generate malicious activity for detection, Hydra was used to perform an FTP brute force attack against Metasploitable2. The SSH attack was attempted first but failed due to cryptographic incompatibility between Hydra's modern algorithms and Metasploitable2's outdated SSH stack (hmac-md5 only). FTP was selected as the attack vector since port 21 was confirmed open and accessible.

```
# SSH attempt failed due to MAC algorithm mismatch
hydra -l msfadmin -P /usr/share/wordlists/rockyou.txt -t 4 ssh://192.168.140.131

# FTP brute force attack (successful)
hydra -l msfadmin -P /usr/share/wordlists/rockyou.txt -t 4 ftp://192.168.140.131
```

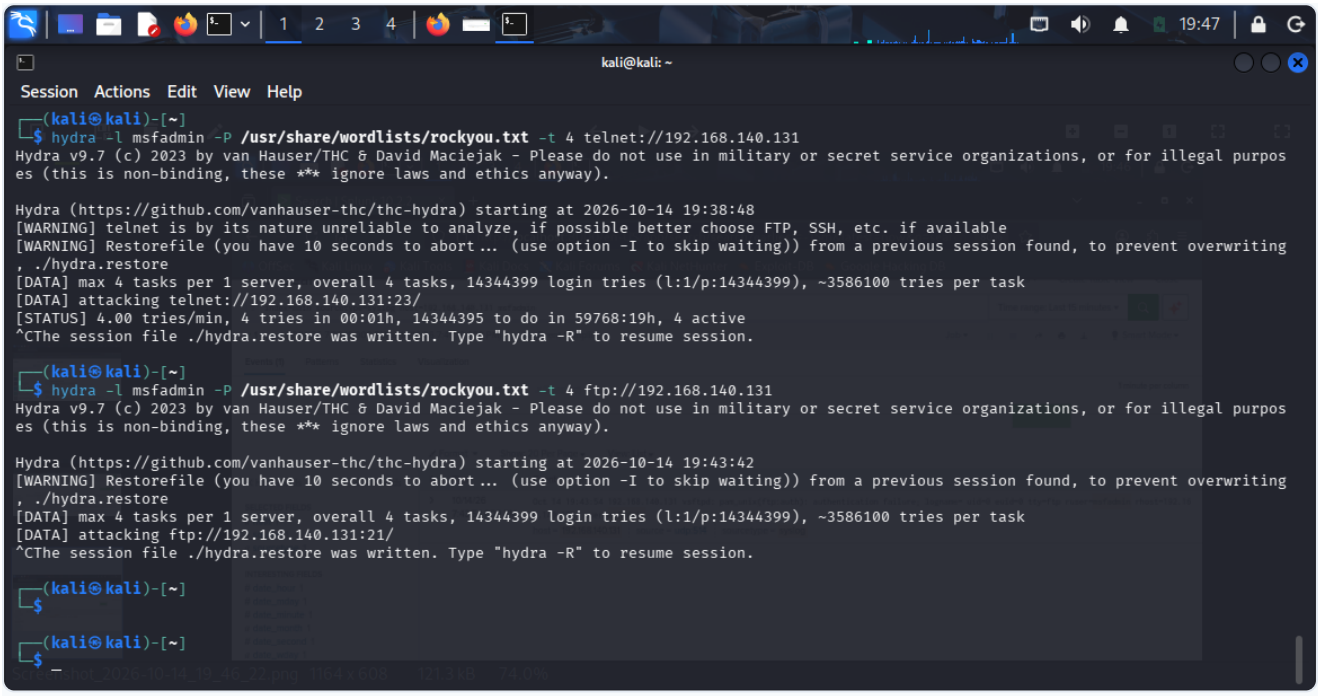


Figure 2: Kali terminal showing Hydra brute force attempts against Telnet and FTP on Metasploitable2

4.2 Detection of Authentication Failures

The FTP brute force attack generated authentication failure logs on Metasploitable2, which were forwarded to Splunk via syslog. The following SPL query was used to detect and visualize these events:

```
index=main sourcetype=syslog host=192.168.140.131 msfadmin
```

The results showed a `vsftpd: pam_unix(ftp:auth): authentication failure` event with the attacker IP `rhost=192.168.140.128` (Kali machine) clearly identified.

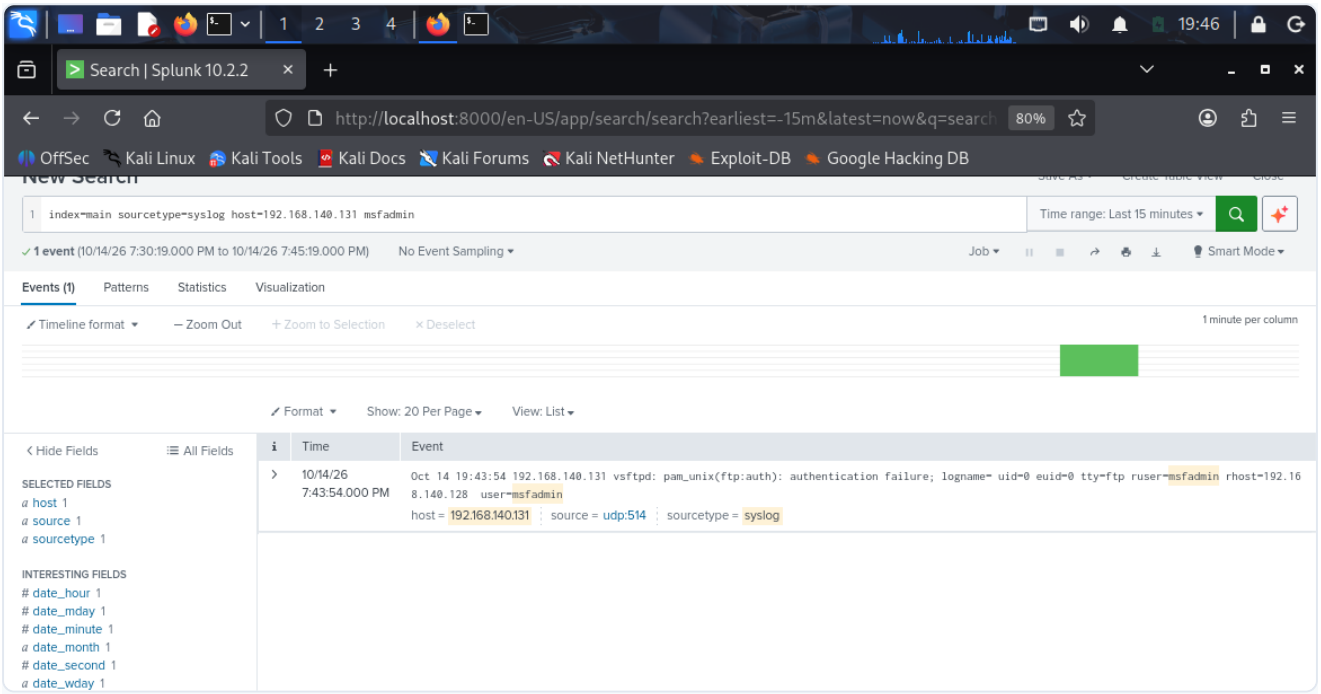


Figure 3: Splunk detection of vsftpd authentication failure from attacker IP 192.168.140.128

4.3 AI-Driven Anomaly Detection Query

A statistical anomaly detection query was built using Splunk SPL. This query extracts the attacker IP using regex, counts authentication failures per source, applies a threshold filter, and assigns a dynamic threat level based on event volume:

```
index=main sourcetype=syslog host=192.168.140.131 "authentication failure"
| rex field=_raw "rhost=(?<attacker_ip>\S+)"
| stats count by attacker_ip
| where count > 0
| eval threat_level=case(count<5,"Low", count<20,"Medium", count⩾20,"High")
| sort - count
```

Query Explanation:

- `index=main sourcetype=syslog` -- Filters to Metasploitable2 syslog events
- `rex field=_raw "rhost=(?<attacker_ip>\S+)"` -- Extracts attacker IP from raw log using regex
- `stats count by attacker_ip` -- Aggregates failure count per source IP
- `where count > 0` -- Displays all IPs with at least one failure
- `eval threat_level=case(...)` -- Assigns Low/Medium/High based on volume thresholds

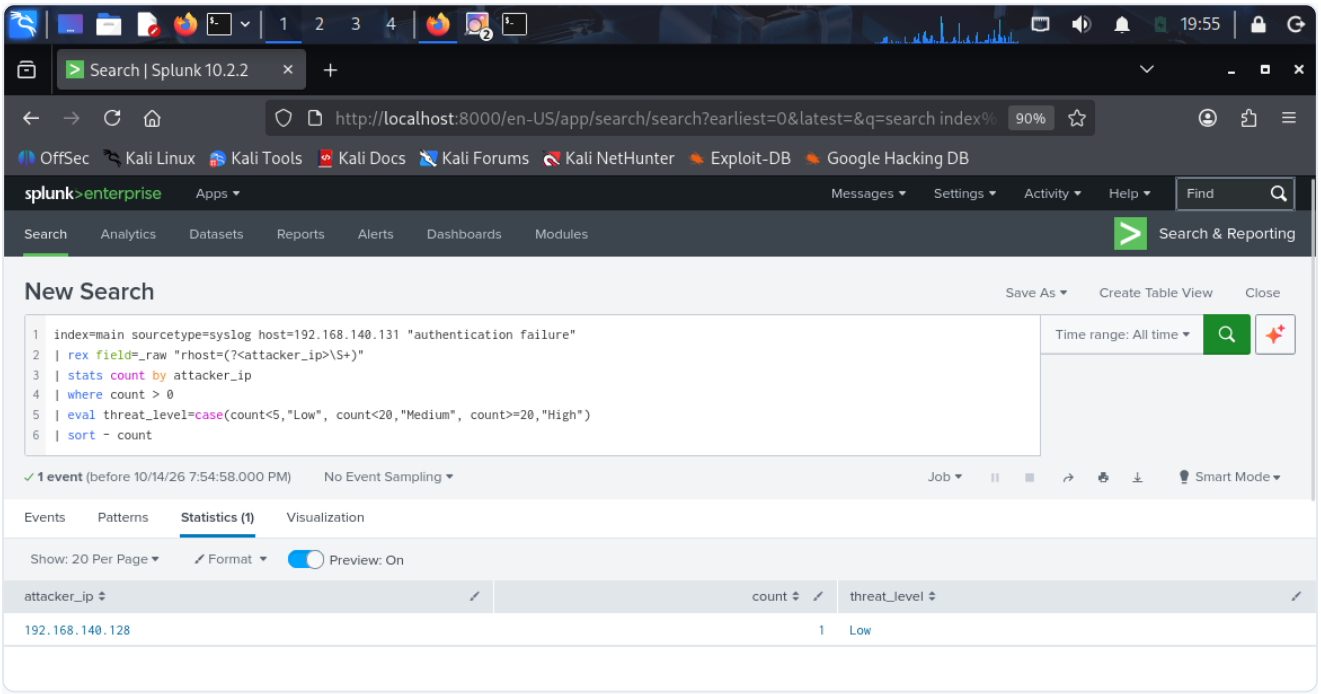


Figure 4: Statistical anomaly detection results showing attacker IP 192.168.140.128 with count=1 and threat_level=Low

4.4 Threat Intelligence Enrichment (Simulated)

In a production environment, OpenCTI would enrich alerts with known threat actor profiles, malware families, and MITRE technique mappings. In this lab, threat intelligence enrichment was simulated using a Splunk lookup table containing the attacker node profile:

```
# threat_intel.csv
ip_address,threat_actor,malware_family,confidence
192.168.140.128,Simulated_Attacker_Node,BruteForce_Tool,High
```

This lookup was registered in Splunk and joined with detection results to demonstrate how IOCs can be automatically enriched with contextual threat intelligence during incident triage.

5. Task 3: Threat Hunting and IOC Pivoting

5.1 IOC Pivot on Attacker IP

Threat hunting was performed by pivoting on the identified attacker IP address (192.168.140.128) to discover all related events across the ingested logs. The following SPL query was used:

```
index=main sourcetype=syslog host=192.168.140.131 192.168.140.128
| sort + _time
| table _time, _raw
```

This query reconstructed the attack timeline by displaying all events involving the attacker IP in chronological order. The timeline revealed:

- syslogd restart (remote reception enabled)
- SPLUNK TEST LOG ENTRY (verification test)
- sudo session activity by msfadmin
- vsftpd authentication failure from rhost=192.168.140.128

5.2 Attack Chain Reconstruction

The complete attack chain observed in the lab was:

1. **Reconnaissance:** Nmap ACK scan (T1046) identified open ports 23 (Telnet) and 80 (HTTP) on Metasploitable2
2. **Initial Access Attempt:** SSH brute force attempted but blocked by cryptographic incompatibility
3. **Credential Access:** FTP brute force (T1110) executed via Hydra against port 21
4. **Detection:** vsftpd authentication failures logged and forwarded to Splunk
5. **Analysis:** Attacker IP extracted via regex, threat level assigned via statistical thresholding

6. Task 4: MITRE ATT&CK and D3FEND Gap Analysis

6.1 MITRE ATT&CK Technique Mapping

Technique ID	Technique Name	Lab Activity	Evidence
T1046	Network Service Scanning	Nmap ACK scan against 192.168.140.131	Nmap output showing ports 23, 80 unfiltered
T1110	Brute Force	Hydra FTP brute force against msfadmin account	vsftpd auth failure logs in Splunk
T1078	Valid Accounts (attempted)	Repeated login attempts using msfadmin username	ruser=msfadmin in syslog events
T1021	Remote Services	FTP (port 21) used as remote access vector	vsftpd pam_unix(ftp:auth) events

6.2 D3FEND Countermeasure Mapping

ATT&CK Technique	D3FEND Countermeasure	Lab Implementation	Status
T1046 (Scanning)	Network Traffic Filtering	Nmap ACK scan mapped firewall rules; no active blocking configured	Partial
T1110 (Brute Force)	Credential Hardening	No account lockout or MFA on Metasploitable2	Gap
T1078 (Valid Accounts)	Identifier Activity Analysis	Splunk detected repeated auth failures from 192.168.140.128	Covered
T1021 (Remote Services)	Network Traffic Filtering	FTP port 21 open with no segmentation or encryption	Gap

6.3 Gap Analysis and Recommendations

Gap Identified	Risk	Recommended Control
No account lockout after repeated failed logins	Brute force attacks can succeed through persistence	Implement account lockout policy (3-5 failed attempts)
FTP (port 21) open with no encryption	Credentials transmitted in plaintext; MITM risk	Disable FTP; migrate to SFTP (port 22) with key-based authentication
No firewall blocking attacker IP	Attacker can retry indefinitely from same source	Deploy host-based firewall (iptables) or network ACLs
No automated response to brute force	Analyst must manually detect and respond	Configure Splunk alerts with automated firewall block actions
No email or web server logs in SIEM	Cannot detect phishing or web-based attacks	Ingest email gateway and web server logs for full coverage

7. Conclusion and Recommendations

7.1 Summary

This project successfully demonstrated the core components of an AI-augmented SOC pipeline. Splunk served as the central SIEM, ingesting syslog data from a Metasploitable2 endpoint. Simulated network reconnaissance (Nmap ACK scanning) and credential-based attacks (Hydra FTP brute force) were detected through Splunk queries with regex-based IOC extraction and statistical anomaly thresholding. The attack chain was mapped to MITRE ATT&CK techniques, and D3FEND gap analysis identified critical deficiencies in credential hardening, remote service exposure, and automated response capabilities.

7.2 Key Learnings

- Syslog forwarding requires proper daemon configuration (`syslogd -r`) and correct syntax (tab-separated fields in `/etc/syslog.conf`)
- Legacy systems like Metasploitable2 often use outdated protocols that modern tools reject, requiring protocol fallback (SSH to FTP)
- Statistical thresholding in SIEM queries provides a foundational AI-driven detection mechanism without requiring enterprise ML platforms

- MITRE ATT&CK and D3FEND frameworks provide a structured vocabulary for communicating attack techniques and defensive gaps to stakeholders

7.3 Future Enhancements

- Deploy OpenCTI with Docker for real-time threat intelligence enrichment instead of simulated lookup tables
- Integrate Shuffle or TheHive for automated incident case creation and response playbook execution
- Install Sophos Intercept X or osquery on endpoints for EDR-grade telemetry beyond syslog
- Implement Splunk MLTK (Machine Learning Toolkit) for unsupervised anomaly detection on authentication patterns
- Expand log sources to include Windows Event Logs, firewall logs, and DNS query logs for multi-vector correlation

8. Appendix: Commands and Configurations

8.1 Splunk Management Commands

```
# Start Splunk
sudo /opt/splunk/bin/splunk start --accept-license --answer-yes --run-as-root

# Check Splunk status
sudo /opt/splunk/bin/splunk status

# Enable Free license (config file method)
sudo tee /opt/splunk/etc/system/local/server.conf > /dev/null << 'EOF'
[license]
active_group = SplunkFree
EOF
```

8.2 Metasploitable2 Syslog Configuration

```
# Edit syslog configuration
sudo nano /etc/syslog.conf

# Add at bottom (use Tab, not spaces):
*.* @192.168.140.128

# Restart syslogd with remote support
sudo killall syslogd
sudo syslogd -r
```

```
# Verify syslogd is running
ps aux | grep syslogd
```

8.3 Attack Simulation Commands

```
# Nmap ACK scan (firewall mapping)
nmap -sA --top-ports 2 192.168.140.131

# FTP brute force with Hydra
hydra -l msfadmin -P /usr/share/wordlists/rockyou.txt -t 4 ftp://192.168.140.131

# Generate test syslog entry
logger "SPLUNK TEST LOG ENTRY"
```

8.4 Splunk Detection Queries

```
# Verify log ingestion
index=main sourcetype=syslog | head 20

# Detect authentication failures
index=main sourcetype=syslog host=192.168.140.131 "authentication failure"

# Statistical anomaly detection with threat level
index=main sourcetype=syslog host=192.168.140.131 "authentication failure"
| rex field=_raw "rhost=(?<attacker_ip>\S+)"
| stats count by attacker_ip
| where count > 0
| eval threat_level=case(count<5,"Low", count<20,"Medium", count≥20,"High")
| sort - count

# IOC pivot and timeline reconstruction
index=main sourcetype=syslog host=192.168.140.131 192.168.140.128
| sort + _time
| table _time, _raw
```